

# CLASS 9 MATH

Advanced Mastery Workbook

*Polynomials, Geometry, Statistics & 100 Olympiad Problems*

# Chapter 1: Number Systems & Polynomials

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## SECTION A: CORE CONCEPTS

1. Rationalize the denominator of  $1 / (\sqrt{7} - \sqrt{6})$ .

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2. Evaluate:  $(64/125)^{-2/3}$ .

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3. Find the remainder when  $x^3 - ax^2 + 6x - a$  is divided by  $x - a$ .

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4. Factorize:  $27x^3 + y^3 + z^3 - 9xyz$ .

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5. Express  $0.2353535\dots$  in the form  $p/q$ .

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## SECTION B: REAL WORLD APPLICATIONS

1. The revenue of the Crunchley makhana brand over 'x' months is modeled by the polynomial  $p(x) = 3x^3 - 5x^2 + 2x - 8$ . Find the revenue at  $x = 3$ .

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2. Mukesh is designing a new algorithm for Jigyasa. The processing time is  $(\sqrt{5} + \sqrt{3}) / (\sqrt{5} - \sqrt{3})$  milliseconds. Rationalize this expression.

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3. A rectangular plot in San Jose has an area of  $(x^2 + 7x + 10)$  square meters. Find the possible expressions for its length and breadth.

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## Chapter 2: Coordinate Geometry & Linear Equations

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### SECTION A: CORE CONCEPTS

1. Find the value of  $k$  if  $x = 2$ ,  $y = 1$  is a solution of the equation  $2x + 3y = k$ .

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2. In which quadrant or on which axis do the points  $(-2, 4)$ ,  $(3, -1)$ ,  $(-1, 0)$ , and  $(-3, -5)$  lie?

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3. Draw the graph of the linear equation  $3x + 4y = 12$ .

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4. Express the linear equation  $2x = 5y$  in the form  $ax + by + c = 0$  and indicate the values of  $a$ ,  $b$ , and  $c$ .

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5. Give the geometric representations of  $y = 3$  as an equation in one variable and in two variables.

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## SECTION B: REAL WORLD APPLICATIONS

1. The cost of a premium Crunchley packet is twice the cost of a standard packet. Write a linear equation in two variables to represent this statement.

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2. Atharv and Anaya together contributed \$50 to a charity. Write a linear equation which satisfies this data and draw its graph.

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3. For the Jigyasa app, the cloud storage cost 'y' is a fixed charge of \$100 plus \$5 per gigabyte 'x' used. Form a linear equation.

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## Chapter 3: Geometry (Triangles & Quadrilaterals)

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### SECTION A: CORE CONCEPTS

1. In a triangle ABC, angle A - angle B =  $33^\circ$  and angle B - angle C =  $18^\circ$ . Find the measure of all three angles.

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2. Prove that the sum of the angles of a quadrilateral is  $360^\circ$ .

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3. If the diagonals of a parallelogram are equal, then show that it is a rectangle.

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4. In triangle ABC, AD is the perpendicular bisector of BC. Show that triangle ABC is an isosceles triangle.

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5. Find the area of a triangle whose sides are 18 cm, 24 cm, and 30 cm using Heron's formula.

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## SECTION B: REAL WORLD APPLICATIONS

1. Priyanka is designing a triangular garden in California. The sides of the triangular garden are in the ratio of 3:5:7 and its perimeter is 300 m. Find its area.

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2. Mukesh maps out a section of a theme park on a coordinate plane. Three rides are at  $A(3, 2)$ ,  $B(3, -2)$ , and  $C(-1, -2)$ . Prove that the triangle formed is a right-angled triangle.

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3. A field is in the shape of a trapezium whose parallel sides are 25 m and 10 m. The non-parallel sides are 14 m and 13 m. Find the area of the field.

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## Chapter 4: Mensuration, Statistics & Probability

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### SECTION A: CORE CONCEPTS

1. Find the volume of a sphere whose radius is 10.5 cm. (Use  $\pi = \frac{22}{7}$ )

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2. The curved surface area of a right circular cylinder of height 14 cm is  $88\text{ cm}^2$ . Find the diameter of the base of the cylinder.

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3. A conical tent is 10 m high and the radius of its base is 24 m. Find the slant height of the tent.

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4. Find the mean of the first 10 prime numbers.

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5. A coin is tossed 1000 times with the following frequencies: Head: 455, Tail: 545. Compute the probability for each event.

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## SECTION B: REAL WORLD APPLICATIONS

1. Priyanka wants to pack Crunchley makhana into cylindrical cans. If a can has a base radius of 7 cm and a height of 20 cm, what is its capacity?

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2. A dome of a building is in the form of a hemisphere. From inside, it was whitewashed at the cost of \$498.96. If the cost of whitewashing is \$2.00 per square meter, find the volume of the air inside the dome.

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3. In a spelling bee assessment for Atharv's class of 30 students, the marks obtained (out of 50) are recorded. If 15 students scored above 40, what is the empirical probability that a student chosen at random scored above 40?

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# Multi-Concept Combined Practice

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## LEVEL 1: CORE PROFICIENCY (AVERAGE STUDENT) - SET 1

1. Rationalize the denominator of  $1 / (\sqrt{5} + \sqrt{2})$ .

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2. Evaluate  $(81)^{(3/4)}$ .

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3. Find the value of the polynomial  $5x - 4x^2 + 3$  at  $x = -1$ .

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4. Check whether  $x + 1$  is a factor of  $x^3 + x^2 + x + 1$ .

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5. Expand using suitable identities:  $(2x - y + z)^2$ .

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6. Find the area of an equilateral triangle with a side of 10 cm.

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7. Write the equation of a line parallel to the x-axis passing through  $(0, -4)$ .

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8. The radius of a spherical balloon increases from 7 cm to 14 cm as air is pumped into it. Find the ratio of their surface areas.

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9. Find the median of the following data: 25, 34, 31, 23, 22, 26, 35, 28, 20, 32.

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10. Two coins are tossed 200 times. Two heads appeared 40 times. Find the probability of getting two heads.

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## Multi-Concept Combined Practice

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### LEVEL 1: CORE PROFICIENCY (AVERAGE STUDENT) - SET 2

1. Simplify:  $(\sqrt{3} + \sqrt{2})^2$ .

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2. Factorize:  $4x^2 - y^2/100$ .

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3. Find the zero of the polynomial  $p(x) = 2x + 5$ .

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4. Plot the points  $(x, y)$  given in the following table on the plane:  $x = [-2, -1, 0, 1]$ ,  $y = [8, 7, -1.25, 3]$ .

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5. If the point  $(3, 4)$  lies on the graph of the equation  $3y = ax + 7$ , find the value of  $a$ .

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6. An angle is equal to its supplement. Find its measure.

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7. The sides of a triangle are 122m, 22m, and 120m. Find its area using Heron's formula.

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8. Find the volume of a right circular cone with radius 6 cm and height 7 cm.

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9. The mean of 5 observations is 15. If the mean of the first 3 is 14 and the last 3 is 17, find the third observation.

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10. In a cricket match, a batswoman hits a boundary 6 times out of 30 balls she plays. Find the probability that she did not hit a boundary.

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## Multi-Concept Combined Practice

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### LEVEL 2: ADVANCED APPLICATION (TOP 10%) - SET 1

1. If  $x = 3 + 2\sqrt{2}$ , find the value of  $x^2 + 1/x^2$ .

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2. Factorize completely:  $x^3 - 23x^2 + 142x - 120$ .

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3. If  $x + y + z = 0$ , show that  $x^3 + y^3 + z^3 = 3xyz$ .

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4. Draw the graphs of the equations  $x - y = 1$  and  $2x + y = 8$ . Shade the area bounded by these two lines and the y-axis.

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5. ABCD is a rhombus. Show that diagonals AC and BD bisect the angles of the rhombus.

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6. In a right triangle ABC, angle B =  $90^\circ$ . If AB = 5 cm and BC = 12 cm, find the radius of the circumcircle of triangle ABC.

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7. A hemispherical bowl made of brass has an inner diameter of 10.5 cm. Find the cost of tin-plating it on the inside at the rate of \$16 per 100 cm<sup>2</sup>.

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8. The mean of 100 observations is 50. If one of the observations which was 50 is replaced by 150, find the new mean.

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9. A die is thrown 1000 times with the frequencies for the outcomes 1, 2, 3, 4, 5, and 6 as 179, 150, 157, 149, 175, and 190. Find the probability of getting an even number.

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10. Simplify:  $(256/81)^{5/4} \times (324/625)^{-1/2}$ .

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## Multi-Concept Combined Practice

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### LEVEL 2: ADVANCED APPLICATION (TOP 10%) - SET 2

1. Express  $1.272727\dots$  as a fraction in simplest form.

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2. Without actually calculating the cubes, find the value of  $(28)^3 + (-15)^3 + (-13)^3$ .

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3. The polynomial  $p(x) = x^4 - 2x^3 + 3x^2 - ax + 3a - 7$  when divided by  $x + 1$  leaves the remainder 19. Find the value of  $a$ .

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4. A taxi fare in a city is as follows: For the first kilometer, the fare is \$8 and for the subsequent distance it is \$5 per km. Taking the distance covered as  $x$  km and total fare as \$ $y$ , write a linear equation and draw its graph.

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5. Prove that angles opposite to equal sides of an isosceles triangle are equal.

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6. The perimeter of an isosceles triangle is 32 cm. The ratio of the equal side to its base is 3:2. Find the area of the triangle.

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7. The capacity of a closed cylindrical vessel of height 1 m is 15.4 liters. How many square meters of metal sheet would be needed to make it?

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8. Find the mode of the following marks (out of 10) obtained by 20 students:  
4, 6, 5, 9, 3, 2, 7, 7, 6, 5, 4, 9, 10, 10, 3, 4, 7, 6, 9, 9.

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9. If  $x^2 + 1/x^2 = 51$ , find the value of  $x - 1/x$ .

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10. Line AB is parallel to CD. A transversal cuts them. If interior angles on the same side of the transversal are  $(2x - 8)^\circ$  and  $(3x - 7)^\circ$ , find the measure of each angle.

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## Multi-Concept Combined Practice

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### LEVEL 3: LOGICAL REASONING (TOP 1%) - SET 1

1. If  $a = (\sqrt{3} - \sqrt{2}) / (\sqrt{3} + \sqrt{2})$  and  $b = (\sqrt{3} + \sqrt{2}) / (\sqrt{3} - \sqrt{2})$ , find the value of  $a^2 + b^2 - 5ab$ .

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2. Factorize:  $x^4 + x^2y^2 + y^4$ .

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3. If  $a + b + c = 5$  and  $ab + bc + ca = 10$ , prove that  $a^3 + b^3 + c^3 - 3abc = -25$ .

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4. Find the coordinates of the vertices of a rectangle placed in the Cartesian plane with its center at the origin, length  $2a$  along the x-axis, and breadth  $2b$  along the y-axis.

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5. In triangle ABC, the bisectors of angle B and angle C intersect at point O. Prove that  $\angle BOC = 90^\circ + \frac{1}{2} \angle A$ .

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6. A kite in the shape of a square with a diagonal 32 cm and an isosceles triangle of base 8 cm and sides 6 cm each is to be made of three different shades. How much paper of each shade has been used in it?

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7. A solid cylinder has a total surface area of  $462 \text{ cm}^2$ . Its curved surface area is one-third of its total surface area. Find the volume of the cylinder.

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8. The mean of  $n$  observations is  $X$ . If the first item is increased by 1, second by 2, and so on, what will be the new mean?

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9. A box contains 50 balls of colors red, green, and blue. The probability of drawing a red ball is  $\frac{2}{5}$  and a green ball is  $\frac{3}{10}$ . Find the number of blue balls in the box.

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10. Solve for x:  $4^x - 10 \cdot 2^{(x-1)} = 24$ .

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## Multi-Concept Combined Practice

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### LEVEL 3: LOGICAL REASONING (TOP 1%) - SET 2

1. Find the value of  $\sqrt{5 + 2\sqrt{6}} + \sqrt{8 - 2\sqrt{15}}$ .

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2. Given that  $x^2 - y^2 = 9$  and  $xy = 3$ , find the value of  $x + y$ .

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3. If the polynomials  $az^3 + 4z^2 + 3z - 4$  and  $z^3 - 4z + a$  leave the same remainder when divided by  $z - 3$ , find the value of  $a$ .

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4. Draw the graph of the equation  $2x + 3y = 12$ . From the graph, find the coordinates of the point whose y-coordinate is 3.

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5. Prove that the line segment joining the mid-points of two sides of a triangle is parallel to the third side and is half of it.

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6. An umbrella is made by stitching 10 triangular pieces of cloth of two different colors, each piece measuring 20 cm, 50 cm, and 50 cm. How much cloth of each color is required for the umbrella?

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7. Water flows at the rate of 10 m/minute through a cylindrical pipe 5 mm in diameter. How long would it take to fill a conical vessel whose diameter at the base is 40 cm and depth 24 cm?

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8. The median of observations 11, 12, 14, 18,  $x+2$ ,  $x+4$ , 30, 32, 35, 41 arranged in ascending order is 24. Find the value of  $x$ .

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9. Evaluate:  $(0.00032)^{(2/5)}$ .

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10. ABCD is a parallelogram and AP and CQ are perpendiculars from vertices A and C on diagonal BD. Show that  $AP = CQ$ .

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## Multi-Concept Combined Practice

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### LEVEL 4: CRITICAL THINKING (TOP 0.1%) - SET 1

1. Simplify:  $\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{8} + \sqrt{9}}$ .

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2. If  $a + b + c = 0$ , find the value of  $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$ .

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3. Let  $P(x)$  be a polynomial of degree 3 such that  $P(1)=1$ ,  $P(2)=2$ ,  $P(3)=3$ , and  $P(4)=5$ . Find  $P(6)$ .

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4. Determine the number of points  $(x, y)$  with integer coordinates that lie strictly inside the triangle formed by the points  $(0,0)$ ,  $(0, 21)$ , and  $(21, 0)$ .

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5. In a right triangle, the length of the hypotenuse is 10, and the perimeter is 24. Find the area of the triangle.

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6. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder. If the height of the cylinder is 10 cm and its base is of radius 3.5 cm, find the total surface area of the article.

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7. There are 5 red shoes, 4 black shoes, and 3 green shoes in a box. You pick out 2 shoes blindly. What is the probability that they form a matching pair?

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8. Find all real solutions to the equation:  $(x^2 - 5x + 5)^{(x^2 - 9x + 20)} = 1$ .

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9. If  $3^x = 4^y = 12^z$ , show that  $z = \frac{xy}{x + y}$ .

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10. A convex polygon has 44 diagonals. How many sides does it have?

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## Multi-Concept Combined Practice

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### LEVEL 4: CRITICAL THINKING (TOP 0.1%) - SET 2

1. Find the exact value of  $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$ .

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2. Factorize:  $x^4 + 4x^3 + 6x^2 + 4x + 1$ .

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3. Two lines intersect at point O. If the sum of three of the angles formed is  $290^\circ$ , find the measure of all four angles.

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4. The area of an equilateral triangle is numerically equal to its perimeter. What is the length of its side?

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5. A right circular cone is divided into two portions by a plane parallel to the base and passing through the midpoint of the height. Find the ratio of the volumes of the two portions.

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6. The mean of 5 positive integers is 5, and their median is 5. If there is only one mode, what is the maximum possible value of the largest integer?

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7. A monkey climbs a 30-meter high greased pole. It goes up 3 meters in one minute and slips down 2 meters in the next minute. In how many minutes will the monkey reach the top?

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8. Find the sum of the digits of the number  $10^{50} - 74$ .

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9. Given the line  $y = mx + c$  passes through (2, 5) and (-1, -4). Find the x-intercept of this line.

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10. If  $x = 1 / (2 - \sqrt{3})$ , find the value of  $x^3 - 2x^2 - 7x + 5$ .

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## Multi-Concept Combined Practice

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### LEVEL 5: NATIONAL OLYMPIAD (TOP 10) - SET 1

1. Find the number of positive integer solutions to the equation  $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$ .

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2. What is the remainder when  $2^{2026}$  is divided by 7?

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3. If  $P(x)$  is a polynomial such that  $P(x^2 - 1) = x^4 - 3x^2 + 3$ , find  $P(x)$ .

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4. In a regular nonagon (9-sided polygon), how many triangles can be formed by joining three of its vertices?

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5. A circle is inscribed in a square of side 10 cm, and another square is inscribed in that circle. Find the area of the smaller square.

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6. You have a  $3 \times 3 \times 3$  cube made of 27  $1 \times 1 \times 1$  cubes. A termite eats its way through the very center of the front face, going straight through to the back face. How many  $1 \times 1 \times 1$  cubes remain completely intact?

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7. How many distinct ways can you arrange the letters in the word 'MATHEMATICS'?

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8. A drawer contains an unorganized mixture of red, blue, green, and yellow socks. You want to guarantee having 4 socks of the exact same color. How many socks must you pull out blindly in the worst-case scenario?

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9. If  $x + y + z = 1$ ,  $x^2 + y^2 + z^2 = 2$ , and  $x^3 + y^3 + z^3 = 3$ , find the value of  $x^4 + y^4 + z^4$ .

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10. Two trains, 500 km apart, travel towards each other at 50 km/h and 75 km/h. A super-fast drone flies back and forth between the trains at 150 km/h until they meet. What is the total distance the drone flew?

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## Multi-Concept Combined Practice

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### LEVEL 5: NATIONAL OLYMPIAD (TOP 10) - SET 2

1. Find the last two digits of the number  $9^{2026}$ .

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2. Let  $f(x) = ax^2 + bx + c$ . If  $f(0) = 1$ ,  $f(1) = 2$ , and  $f(2) = 5$ , what is the value of  $f(3)$ ?

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3. In triangle ABC, a point D is on BC such that  $BD:DC = 1:2$ , and E is on AD such that  $AE:ED = 2:3$ . Find the ratio of the area of triangle ABE to the area of triangle ABC.

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4. What is the greatest number of pieces a spherical cheese ball can be cut into with exactly 5 straight planar cuts?

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5. A water tank has three pipes. Pipe A fills it in 3 hours, Pipe B fills it in 4 hours, and Pipe C empties it in 6 hours. If they are opened in order at 8 AM, 9 AM, and 10 AM respectively, when will the tank be full?

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6. Find the sum of all irreducible fractions with a denominator of 3 that are between 10 and 20.

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7. A sequence is defined by  $a_1 = 1$ ,  $a_2 = 1$ , and  $a_n = a_{(n-1)} + a_{(n-2)}$  for  $n > 2$ . What is the greatest common divisor of  $a_{100}$  and  $a_{101}$ ?

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8. If the roots of the equation  $x^3 - 12x^2 + 39x - 28 = 0$  are in arithmetic progression, find the roots.

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9. A spherical drop of water loses its volume by evaporation at a rate proportional to its surface area. If it loses half its volume in 1 hour, how long will it take to evaporate completely?

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10. How many trailing zeros are at the end of the number  $1000!$  (1000 factorial)?

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